DATABASE DOCUMENTATION FOR

LIBRARY MANAGEMENT SYSTEM

In Partial Fulfillment of the Requirements

for CC 105 – Information Management and CC106 – Application Development and Emerging Technologies

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2026

INTRODUCTION

This paper presents the database documentation for the Library Management System developed by Group 6: Reneverse. The system uses a MySQL database named library_management to organize information about books, library branches, members, and borrowing transactions. The database also includes the users and book_requests tables, which support the login and book-request features implemented in the website/application.

The documentation discusses the database structure, relationships, data dictionary, normalization, and sample SQL queries. The Entity Relationship Diagram (ERD) and normalization discussion are limited to the four main tables: books, book_loans, members, and branches. The users and book_requests tables are discussed in the other relevant sections because they are used by the website/application developed by Group 6: Reneverse.

OBJECTIVE

- To store and manage book information, including ISBN, title, author, publisher, classification, location, and availability.
- To maintain library branch details and associate books, members, loans, and requests with relevant branches.
- To maintain member information and membership status.
- To record borrowing transactions, due dates, returns, loan status, renewals, and late-fine amounts.
- To support user registration, login, and role-based access through the users table.
- To record and manage book requests submitted by system users.
- To demonstrate CRUD operations, relational joins, filtering, aggregation, and reporting using SQL.

DATABASE OVERVIEW

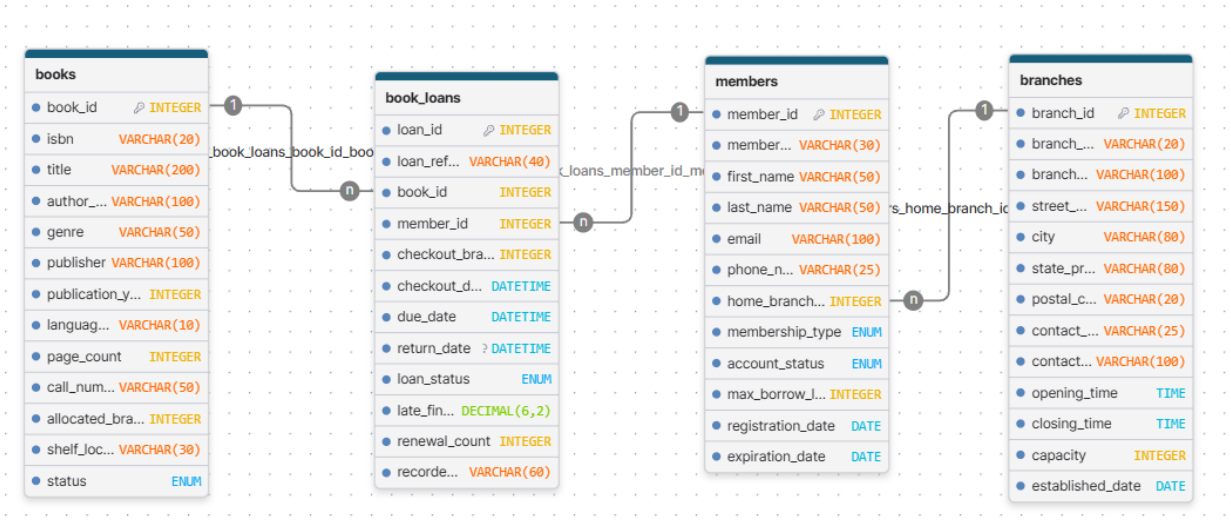
The library_management database consists of six tables: books, book_loans, branches, members, users, and book_requests. The books, book_loans, branches, and members tables represent the main library records. The users and book_requests tables support the website/application by storing account information and submitted borrowing requests.

	Purpose
Books	Stores book information, classification, branch allocation, shelf location, and status.
book_loans	Stores borrowing transactions, due/return dates, loan status, and late fines.
branches	Stores library branch details and contact information.
members	Stores library member and membership information.
users	Stores website/application login accounts, usernames, password hashes, and roles.
book_requests	Stores users' book requests, selected pickup branch, request status, and review details.

Note: The users and book_requests tables are used in the website/application developed by the Group 6: Reneverse. The users table stores account details required for login and role-based access, while the book_requests table stores borrowing requests submitted through the application.

Entity Relationship Diagram (ERD)

The ERD presented in this section illustrates the relationships among books, book_loans, members, and branches. The users and book_requests tables are not included in the ERD.



This diagram shows how the library records are related. Books may be assigned to a branch, members may be associated with a home branch, and each book loan record connects a member with a borrowed book. The checkout branch is also recorded to identify the branch associated with the borrowing transaction.

DATABASE RELATIONSHIPS

The database uses primary keys and foreign keys to connect records across tables.

Related Tables	**Relationship Type**	**Primary / Foreign Key Link**	**Description**
Branches — Books	One-to-many (1:M)	branches.branch_id → books.allocated_branch_id	One branch can have many books assigned to it. Each book may be assigned to a branch.
Branches — Members	One-to-many (1:M)	branches.branch_id → members.home_branch_id	One branch can be the home branch of many members. A member's home branch is stored in home_branch_id.
Books — Book Loans	One-to-many (1:M)	books.book_id → book_loans.book_id	One book can appear in several loan records over time. Each loan record refers to a book.
Members — Book Loans	One-to-many (1:M)	members.member_id → book_loans.member_id	One member can have multiple borrowing records. Each loan record identifies the member who borrowed the book.
Branches — Book Loans	One-to-many (1:M)	branches.branch_id → book_loans.checkout_branch_id	One branch can be recorded for many checkout transactions. This field identifies the branch where the loan was processed.
Users — Book Requests (request owner)	One-to-many (1:M)	users.user_id → book_requests.user_id	One user account can submit multiple book requests. The user_id field identifies who submitted each request.
Books — Book Requests	One-to-many (1:M)	books.book_id → book_requests.book_id	One book can be referenced by multiple requests. Each request

			identifies the book being requested.
Branches — Book Requests	One-to-many (1:M)	branches.branch_id → book_requests.pickup_branch_id	One branch can be selected as the pickup location for many requests. The field stores the requested pickup branch.
Users — Book Requests (reviewer)	One-to-many (1:M)	users.user_id → book_requests.reviewed_by	One user account can review multiple requests. reviewed_by may be NULL until a request has been reviewed.

Note: The SQL schema does not define a direct foreign-key relationship between users and members. They serve different purposes in the current database: users is for application login, while members store library membership records if the users borrow a book (in application).

DATA DICTIONARY

The data dictionary lists the fields, data types, and key purposes based on the provided SQL schema. Enum fields are shown with their permitted values where available.

1. books Table

Field	**Data type**	**Key / purpose**
book_id	INT	Primary key; unique book record ID
isbn	VARCHAR(20)	Unique ISBN
title	VARCHAR(200)	Book title
author_name	VARCHAR(100)	Author name
genre	VARCHAR(50)	Book genre/category
publisher	VARCHAR(100)	Publisher name
publication_year	INT	Publication year
language_code	VARCHAR(10)	Language code; default ENG
page_count	INT	Number of pages
call_number	VARCHAR(50)	Library classification/call number

allocated_branch_id	INT	Foreign key to branches.branch_id
shelf_location	VARCHAR(30)	Physical shelf location
status	ENUM('AVAILABLE','CHECKED OUT','RESERVED','UNDER MAINTENANCE','LOST')	Available, Checked Out, Reserved, Under Maintenance, or Lost

2. book_loans Table

Field	**Data type**	**Key / purpose**
loan_id	INT	Primary key; unique loan ID
loan_reference	VARCHAR(40)	Unique loan reference
book_id	INT	Foreign key to books.book_id
member_id	INT	Foreign key to members.member_id
checkout_branch_id	INT	Foreign key to branches.branch_id
checkout_date	DATETIME	Date and time of checkout
due_date	DATETIME	Due date and time
return_date	DATETIME	Return date/time; may be NULL
loan_status	ENUM	Ongoing, Returned, Overdue, or Lost
late_fine_amount	DECIMAL(6,2)	Late fine amount
renewal_count	INT	Number of renewals
recorded_by_staff	VARCHAR(60)	Staff identifier/name recorded with the loan

3. branches Table

Field	Data type	Key / purpose
branch_id	INT	Primary key; branch ID
branch_code	VARCHAR(20)	Unique branch code
branch_name	VARCHAR(100)	Branch name
street_address	VARCHAR(150)	Street address
city	VARCHAR(80)	City
state_province	VARCHAR(80)	State/province
postal_code	VARCHAR(20)	Postal code
contact_phone	VARCHAR(25)	Branch phone
contact_email	VARCHAR(100)	Branch email
opening_time / closing_time	TIME	Operating hours
capacity	INT	Branch capacity
established_date	DATE	Date branch was established

4. members Table

Field	Data type	Key / purpose
member_id	INT	Primary key; member ID
membership_number	VARCHAR(30)	Unique membership number
first_name / last_name	VARCHAR(50)	Member name
email	VARCHAR(100)	Unique member email
phone_number	VARCHAR(25)	Member phone number
home_branch_id	INT	Foreign key to branches.branch_id
membership_type	ENUM	Student, Faculty, Regular, Senior, or Corporate
account_status	ENUM	Active, Suspended, Expired, or Pending
max_borrow_limit	INT	Maximum number of books allowed
registration_date / expiration_date	DATE	Membership dates

5. users Table

Supports login/sign-up and role-based access for the website/application.

Field	Data type	Key / purpose
user_id	INT UNSIGNED	Primary key; account ID
first_name / last_name	VARCHAR(50)	Account holder name
email	VARCHAR(100)	Unique email
username	VARCHAR(50)	Unique username
password	VARCHAR(255)	Password value stored by the application (hash)
role	ENUM('ADMIN','USER')	Admin or User; default User

6. book_requests Table

Stores borrowing requests created and reviewed through the website/application.

Field	Data type	Key / purpose
request_id	INT UNSIGNED	Primary key; primary key; unique request id
user_id	INT UNSIGNED	Foreign key to users.user_id
book_id	INT	Foreign key to books.book_id
pickup_branch_id	INT	Foreign key to branches.branch_id
request_note	VARCHAR(500)	Request note; defaults to empty string
status	ENUM('PENDING','APPROVED','REJECTED','CANCELLED')	Pending, Approved, Rejected, or Cancelled;

		defaults to Pending
request_date	DATETIME	Date/time the request was created
reviewed_by	INT UNSIGNED	Foreign key to users.user_id; nullable until reviewed
reviewed_at	DATETIME	Date/time the request was reviewed; may be NULL

DATABASE NORMALIZATION

Database normalization is the process of organizing data in a database to reduce unnecessary data duplication, maintain data consistency, and establish proper relationships between tables. It helps ensure information is stored efficiently and that changes can be managed without creating unnecessary inconsistencies.

The Library Management System database uses four main tables for this normalization analysis: books, branches, members, and book_loans. These tables use primary keys, unique keys, and foreign keys to identify records and establish relationships between related information.

FIRST NORMAL FORM (1NF)

First Normal Form (1NF) requires that each table has a primary key and that each field contains a single, atomic value rather than a repeating group or multiple values stored in one field.

In the Library Management System, the four main tables use individual columns for their attributes. For example, the books table stores the book title, author name, genre, publisher, publication year, and page count in separate fields. The members table also stores member information in separate columns, such as first name, last name, email, and phone number.

The book_loans table stores loan information, including the book ID, member ID, checkout date, due date, and loan status, in separate fields.

SECOND NORMAL FORM (2NF)

Second Normal Form (2NF) requires that a table is already in 1NF and that every non-key attribute depends on the whole candidate key, rather than only part of a composite key.

The main tables use individual primary keys:

- books — book_id
- branches — branch_id
- members — member_id
- book_loans — loan_id

These primary keys uniquely identify records in their respective tables. Because the shown tables use single-column primary keys, partial dependencies on part of a composite primary key do not apply to those primary keys.

THIRD NORMAL FORM (3NF)

Third Normal Form (3NF) requires that a table is already in 2NF and that non-key attributes do not depend transitively on a candidate key. In other words, non-key attributes should depend on the key, the whole key, and nothing but the key.

The Library Management System separates branch information into the branches table and references branch records through foreign keys in other tables.

For example:

- The books table uses allocated_branch_id to reference a branch.
- The members table uses home_branch_id to reference a branch.
- The book_loans table uses checkout_branch_id to reference a branch.

This structure allows branch information to be maintained in the branches table instead of repeating the complete branch details in every related book, member, or loan record.

Similarly, the book_loans table references the related book and member through book_id and member_id. This establishes relationships without requiring all book and member details to be duplicated in each loan record**.**

ANALYSIS OF THE MAIN DATABASE TABLES

Table	**Primary Key**	**Important Unique key or FKs**	**Purpose**
books	book_id	isbn (unique); allocated_branch_id (FK)	Stores book information and its allocated branch.
branches	branch_id	branch_code (unique)	Stores branch information, including address and contact details.
Members	member_id	email (unique); home_branch_id (FK)	Stores member information and the member's home branch.
Book_loans	loan_id	loan_reference (unique); book_id, member_id, and checkout_branch_id (FKs)	Stores loan transactions and links them to books, members, and branches.

Note: *FK means foreign key. The unique constraints shown in the screenshots help prevent duplicate values in the corresponding columns.*

EVIDENCE OF DATABASE STRUCTURE AND RELATIONSHIPS

The following screenshots support the database structure, keys, and relationships implemented in the Library Management System.

1. Books Table Structure

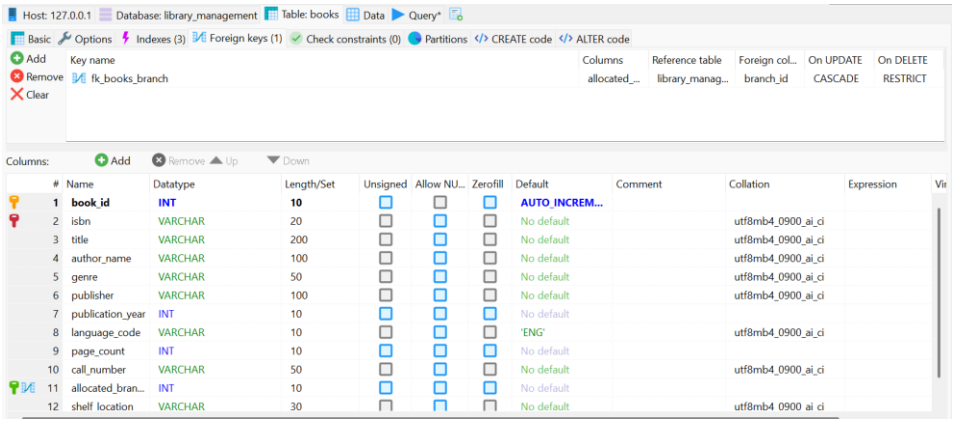


Figure 1. *Structure of the books table showing the book_id primary key, the unique isbn field, and the foreign key allocated_branch_id referencing the branches table.*

The screenshot shows that each book has a primary identifier and a unique ISBN constraint. It also shows the foreign key used to associate a book with a branch.

2. Branches Table Structure

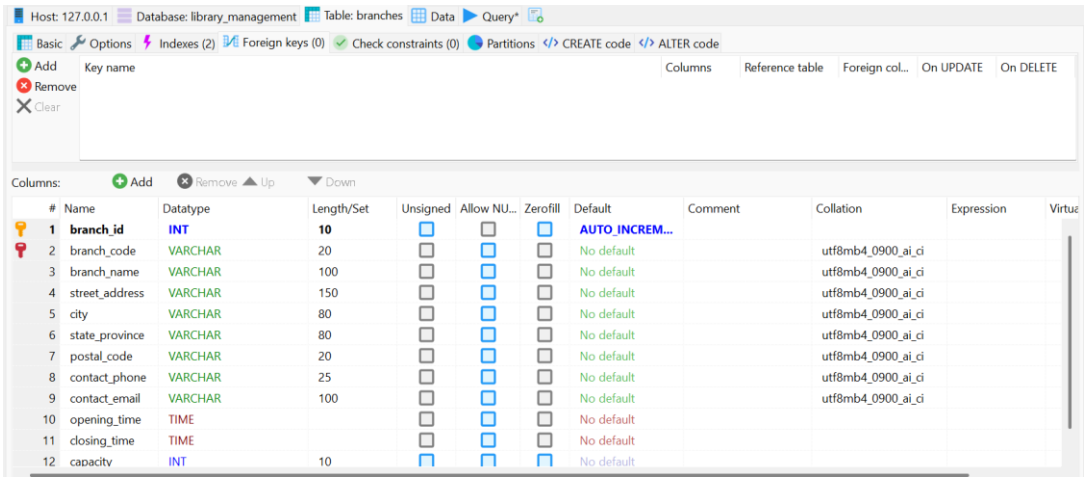


Figure 2. *Structure of the branches table showing branch_id as the primary key and branch_code as a unique field.*

3. Members Table Structures

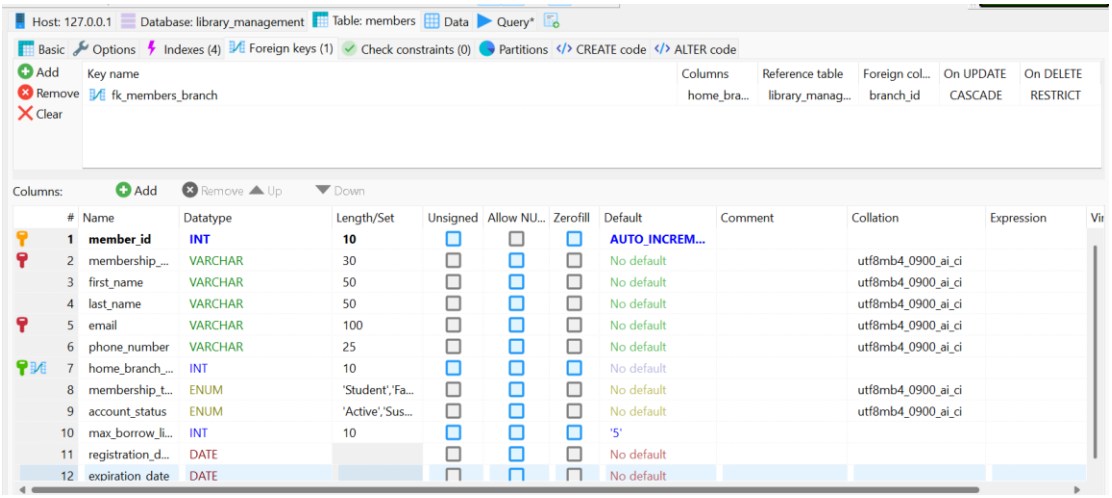


Figure 3. *Structure of the members table showing member_id as the primary key, email as a unique field, and home_branch_id as a foreign key referencing the branches table.*

The screenshot demonstrates how member information is stored separately from branch information. The foreign key establishes a relationship between a member and the member's home branch.

4. Book Loans Table Structure

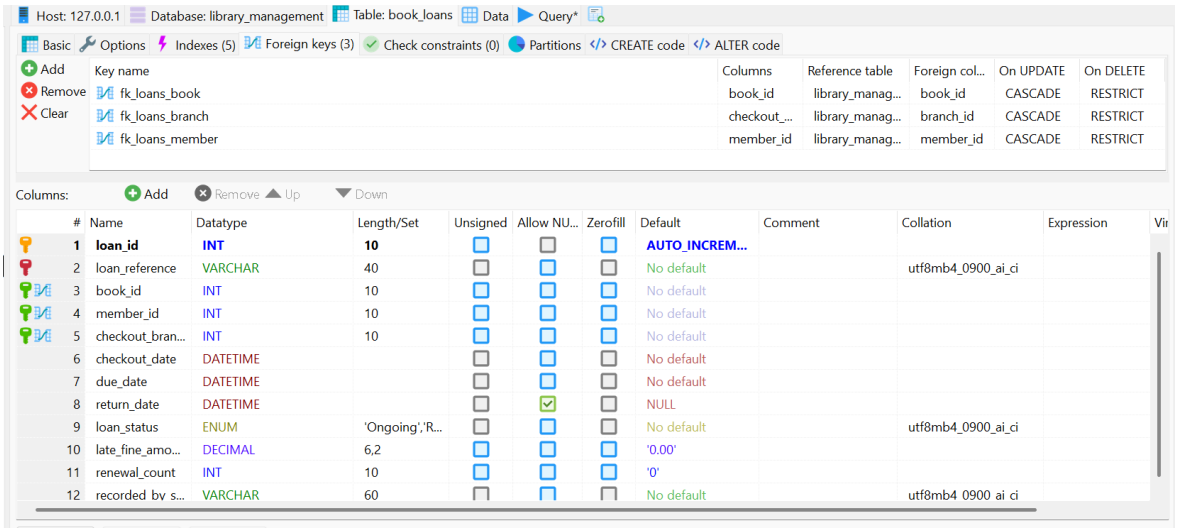


Figure 4. *Structure of the book_loans table showing loan_id as the primary key, loan_reference as a unique field, and foreign keys for the related book, member, and checkout branch.*

The screenshot shows that the book_loans table records borrowing transactions and connects them to the corresponding records in the books, members, and branches tables. These foreign key relationships help maintain referential integrity**.**

Summary of Database Relationships

The main relationships shown by the foreign keys are:

1. Branches and Books: A branch may be associated with multiple books through books.allocated_branch_id.
2. Branches and Members: A branch may be associated with multiple members through members.home_branch_id.
3. Books and Book Loans: A book may be referenced by loan records through book_loans.book_id.
4. Members and Book Loans: A member may be referenced by loan records through book_loans.member_id.
5. Branches and Book Loans: A branch may be referenced by loan records through book_loans.checkout_branch_id.

These relationships help the database connect related information while keeping the main entities in separate tables.

CONCLUSION

The Library Management System organizes library information through related tables for books, members, branches, borrowing transactions, and system users. Primary keys and foreign keys help maintain relationships between records, while SQL queries support searching, reporting, and monitoring of library activities. Normalization principles help reduce unnecessary duplication and improve data consistency. With a properly tested schema, sample data, and documented query results, the system can serve as a structured database solution for library operations.

SQL QUERIES AND DATABASE OPERATIONS DOCUMENTATION

This documentation presents SQL queries used in the Library Management System. It demonstrates how to insert, update, and delete book records, retrieve information, summarize data using aggregate functions, connect related tables using JOIN, analyze categorical and financial data, and filter grouped results using HAVING. The queries are intended for a MySQL database. Table names, column names, and sample values should match the actual database structure.

Data Manipulation Language (DML)

DML commands allow users to add, modify, and remove records in database tables.

1. **INSERT — Add a Book**

Purpose: Adds a new book record to the books table.

```sql
INSERT INTO books (
    isbn, title, author_name, genre, publisher,
    publication_year, language_code, page_count,
    call_number, allocated_branch_id, shelf_location, status
)
VALUES (
    'BBM-Method-1123',
    'BBM & INDAY FRIENDSHIP STORY',
    'Roland Baesa',
    'Fantasy',
    'The OVP',
    2026,
    'FILI',
    300,
    '+6399999999999',
    1,
    'LIKOD NG BAHAY',
    'Available'
);
```

The **INSERT INTO** statement adds a new record to the books table. The specified **values** correspond to the listed columns.

2. **UPDATE — Modify a Book**

Purpose: Changes the genre and status of a specific book.

```
UPDATE books
SET
    genre = 'Adventure',
    status = 'Lost'
WHERE book_id = 1039;
```

The **UPDATE** statement modifies an existing record. The **SET** clause specifies the new values, while the **WHERE** clause identifies the book to modify.

3. **DELETE — Remove a Book**

Purpose: Removes a specific book record from the books table.

```
DELETE FROM books
WHERE book_id = 1037;
```

The **DELETE** statement removes the record whose **book_id** is **1037.**

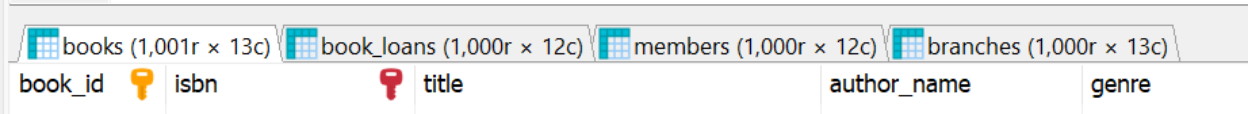
BASIC SELECT QUERIES

SELECT queries retrieve information from database tables.

1. Display All Records (SELECT *)

Purpose: Displays all columns and records from the selected table.

```
1  SELECT * FROM books;
2
3  SELECT * FROM book_loans;
4
5  SELECT * FROM members;
6
7  SELECT * FROM branches;
```



The **asterisk (*)** represents all columns. Each statement displays the contents of one table.

2. Display Specific Columns

Purpose: Displays only the title, author, and genre of each book.

```
1  SELECT title, author_name, genre
2  FROM books;
```

books (1,001r × 3c)

title	author_name	genre
Mastering Data Structures Edition...	Prof. Grace Hopper	Mathematics
The Foundations of Artificial Inte...	Dennis Ritchie	History
Applied Cybersecurity Edition 4	Ada Lovelace	Natural Science

This query retrieves only the specified columns instead of displaying every column in the table.

3. Filter Books by Status

Purpose: Displays books whose status is Available.

```
2  FROM books
3  WHERE status = 'Available';
```

books (397r × 3c)

title	author_name	status
The Foundations of Artificial Inte...	Dennis Ritchie	Available
Essential Guide to Linear Algebra...	Donald Knuth	Available

The **WHERE** clause filters the results and displays only books that match the specified status**.**

4. Sort Books Alphabetically

Purpose: Arranges book titles in ascending alphabetical order.

```
2  FROM books
3  ORDER BY title ASC;
```

books (1,001r × 3c)

title	author_name	genre
Advanced Topics in World Histor...	Linus Torvalds	Philosophy
Advanced Topics in World Histor...	Linus Torvalds	Computer Science
Advanced Topics in World Histor...	Linus Torvalds	Computer Science

ORDER BY sorts the results**. ASC** means ascending order A to Z, while **DESC** Z to A

Basic Filtered Query (SELECT, WHERE, ORDER BY)

Purpose: Displays available books and sorts them alphabetically by title.

```
1  SELECT title, author_name, genre, status
2  FROM books
3  WHERE status = 'Available'
4  ORDER BY title ASC;
```

books (397r × 4c)

title	author_name	genre	status
Advanced Topics in World Histor...	Linus Torvalds	Computer Science	Available
Advanced Topics in World Histor...	Linus Torvalds	Philosophy	Available
Advanced Topics in World Histor...	Linus Torvalds	Philosophy	Available
Advanced Topics in World Histor...	Linus Torvalds	Computer Science	Available

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Aggregated Summary (GROUP BY, COUNT, AVG)

Aggregate functions summarize multiple records and return calculated results.

1. **GROUP BY — List Membership Types**

Purpose: Displays the different membership types recorded in the members table.

```
1  SELECT membership_type
2  FROM members
3  GROUP BY membership_type;
```

members (5r × 1c)

membership_type
Faculty
Regular
Senior
Corporate
Student

GROUP BY groups records with the same membership type. In this query, it returns one row for each distinct membership type.

2. **COUNT() — Count All Books**

Purpose: Calculates the total number of records in the books table.

```
1  SELECT COUNT(*) AS total_books
2  FROM books;
```

books (1r × 1c)

total_books
1,001

COUNT(*) counts the rows in the table. The alias **total_books** gives the calculated result a readable column name.

3. **AVG() — Calculate Average Page Count**

Purpose: Calculates the average number of pages across the books

```
1  SELECT AVG(page_count) AS average_pages
2  FROM books;
```

books (1r × 1c)

average_pages
494.7053

AVG() calculates the average of the page_count values.

Aggregated Summary by Genre

Purpose: Groups books by genre, counts the books in each genre, and calculates their average page count.

```
1  SELECT
2      genre,
3      COUNT(*) AS total_books,
4      ROUND(AVG(page_count), 2) AS average_pages
5  FROM books
6  GROUP BY genre;
```

books (6r × 3c)

genre	total_books	average_pages
Mathematics	167	487.9
History	167	500.9
Natural Science	167	513.9
Philosophy	168	499.36
Engineering	166	489.16
Computer Science	166	476.86

- **GROUP BY** genre groups books according to genre.
- **COUNT(*)** calculates the number of books per genre.
- **AVG(page_count)** calculates the average page count.
- **ROUND(..., 2)** rounds the average to two decimal places.

This query provides a summary of the number of books and their average page counts for each genre.

```sql
SELECT
    genre,
    COUNT(*) AS total_books,
    ROUND(AVG(page_count), 2) AS average_pages
FROM books
GROUP BY genre
ORDER BY total_books DESC;
```

books (6r × 3c)

genre	total_books	average_pages
Philosophy	168	499.36
Mathematics	167	487.9
History	167	500.9
Natural Science	167	513.9
Engineering	166	489.16
Computer Science	166	476.86

The query performs the same aggregation as the previous example but adds **ORDER BY total_books DESC**. **DESC** sorts the results in descending order (highest to lowest)

Multi-Table Relationship (JOIN)

JOIN queries retrieve related information from two or more tables through matching columns, commonly primary keys and foreign keys.

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1. **JOIN Between Books and Branches**

Purpose: Displays each book alongside the name of its allocated library branch.

```sql
SELECT
    b.title,
    b.author_name,
    br.branch_name
FROM books AS b
INNER JOIN branches AS br
    ON b.allocated_branch_id = br.branch_id;
```

Result #1 (1,001r × 3c)

title	author_name	branch_name
Introduction to Database System...	Dr. Alan Turing	East Wing Public Library 1
Applied Cybersecurity Edition 3	Ada Lovelace	North Campus Public Library 2

The query connects books and branches through **allocated_branch_id** and **branch_id.**

- **b** is an alias for **books.**
- **br** is an alias for **branches.**
- **INNER JOIN** returns records with matching **branch IDs.**

This query demonstrates a relationship between two database tables.

2. **JOIN Between Members, Book Loans, and Books**

Purpose: Displays the member's name, borrowed book title, and loan status.

```sql
SELECT
    m.first_name,
    m.last_name,
    b.title,
    bl.loan_status
FROM members AS m
INNER JOIN book_loans AS bl
    ON m.member_id = bl.member_id
INNER JOIN books AS b
    ON bl.book_id = b.book_id;
```

Result #1 (1,000r × 4c)

first_name	last_name	title	loan_status
Paolo	Aquino	Mastering Data Structures Edition...	Returned
Ana	Bautista	The Foundations of Artificial Inte...	Returned

This query connects three tables:

- members provides the member's name.
- book_loans provides the loan record and loan status.
- books provides the title of the associated book.

The relationships are established through member_id and book_id.

3. JOIN Between Members, Books, Book Loans, and Branches

Purpose: Displays member information, book titles, loan status, and allocated branch names in one result.

```sql
SELECT
    m.first_name,
    m.last_name,
    b.title AS book_title,
    bl.loan_status AS loan_status,
    br.branch_name
FROM members AS m
INNER JOIN book_loans AS bl
    ON m.member_id = bl.member_id
INNER JOIN books AS b
    ON bl.book_id = b.book_id
INNER JOIN branches AS br
    ON b.allocated_branch_id = br.branch_id;
```

Result #1 (1,000r × 5c)

first_name	last_name	book_title	loan_status	branch_name
Maria	Reyes	Introduction to Database System...	Ongoing	East Wing Public Library 1
Gabriel	Ramos	Applied Cybersecurity Edition 3	Overdue	North Campus Public Library 2

This query combines four related tables. It shows how **JOIN** can retrieve connected information without duplicating the details of **members, books, and branches** in a single table. Because it uses **INNER JOIN**, records without matching relationships are excluded from the results.

Financial and Categorical Breakdown (SUM, ROUND)

This section demonstrates how SQL can summarize numerical information. A categorical breakdown groups data by categories, while a financial breakdown summarizes monetary values.

1. SUM() — Calculate Total Pages

Purpose: Calculates the total number of pages across all books with recorded page counts.

```sql
SELECT SUM(page_count) AS total_pages
FROM books;
```

Result #1 (1,000r × 5c)

first_name	last_name	book_title	loan_status	branch_name
Maria	Reyes	Introduction to Database System...	Ongoing	East Wing Public Library 1
Gabriel	Ramos	Applied Cybersecurity Edition 3	Overdue	North Campus Public Library 2

SUM() adds the numerical values in the page_count column.

2. ROUND() — Calculate Average Page Count

Purpose: Displays the average page count rounded to two decimal places.

```
1  SELECT ROUND(AVG(page_count), 2) AS average_pages
2  FROM books;
```

books (1r × 1c)

average_pages
494.71

AVG() calculates the average, while **ROUND(..., 2)** limits the displayed result to two decimal places.

3. Categorical Breakdown by Genre

Purpose: Calculates the total pages and average page count for each genre, then sorts the genres from the highest to the lowest average page count.

```
1  SELECT
2      genre,
3      SUM(page_count) AS total_pages,
4      ROUND(AVG(page_count), 2) AS average_pages
5  FROM books
6  GROUP BY genre
7  ORDER BY average_pages DESC;
```

books (6r × 3c)

genre	total_pages	average_pages
Natural Science	85,821	513.9
History	83,650	500.9
Philosophy	83,892	499.36
Engineering	81,200	489.16
Mathematics	81,479	487.9
Computer Science	79,158	476.86

The query groups books by genre and calculates two measurements for each group. **SUM(page_count)** calculates the total pages, and **ROUND(AVG(page_count), 2)** calculates the rounded average.This is a categorical breakdown because the results are organized by book genre.

4. **Financial Breakdown by Loan Status**

Purpose: Calculates the total late fines for each loan status and sorts the results from the highest total fines to the lowest.

```sql
SELECT
    loan_status,
    ROUND(SUM(late_fine_amount), 2) AS total_fines
FROM book_loans
GROUP BY loan_status
ORDER BY total_fines DESC;
```

book_loans (3r × 2c)

loan_status	total_fines
Overdue	62,500.0
Returned	25,000.0
Ongoing	0.0

The query groups loan records according to loan_status. It then calculates the total fines for each group and rounds the result to two decimal places. This is a financial breakdown because it summarizes monetary amounts.

Conditional Aggregation (HAVING)

The HAVING clause filters groups created by GROUP BY. Unlike WHERE, which filters individual rows before grouping, HAVING filters the aggregated results.

1. Count Books per Genre

Purpose: Calculates the number of books in each genre.

```
1  SELECT
2      genre,
3      COUNT(*) AS total_books
4  FROM books
5  GROUP BY genre;
```

books (6r × 2c)

genre	total_books
Mathematics	167
History	167
Natural Science	167
Philosophy	168
Engineering	166
Computer Science	166

GROUP BY genre creates a group for each genre, and **COUNT(*)** calculates the number of books in each group. This query establishes the grouped totals before applying a condition.

2. Display Genres with Exactly 166 Books

Purpose: Displays only the genres that contain exactly 166 book records.

```
1  SELECT
2      genre,
3      COUNT(*) AS total_books
4  FROM books
5  GROUP BY genre
6  HAVING COUNT(*) = 166;
```

books (2r × 2c)

genre	total_books
Engineering	166
Computer Science	166

HAVING COUNT(*) = 166 filters the grouped results. Only genres with exactly 166 books appear in the output. If no genre contains exactly 166 books, the query returns no rows.

3. Display Genres with Exactly 167 Books

Purpose: Displays only the genres that contain exactly 167 book records.

```sql
SELECT
    genre,
    COUNT(*) AS total_books
FROM books
GROUP BY genre
HAVING COUNT(*) = 167;
```

books (3r × 2c)

genre	total_books
Mathematics	167
History	167
Natural Science	167

This query works like the previous example, but the condition requires exactly 167 books per genre. The values 166 and 167 are alternative conditions, not requirements that must both be used. Choose a count that is appropriate for your actual data or use them to demonstrate how HAVING works.